Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

1 The decision

An indie studio's creative director chooses the two tags for the next game. If no pair passed the selection rule (section 5), they would take a proven but crowded pair such as Building × Simulation.

2 Why a network

Counts can't show whether two tags attract different players. The network can: tags with few shared neighbours point to audiences that rarely meet, which one game may reach together.

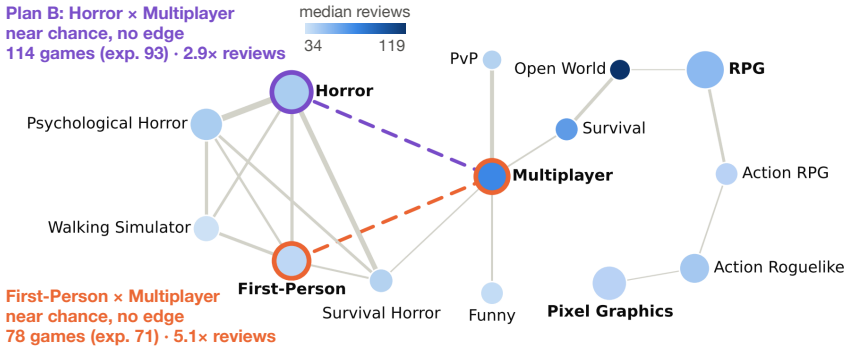
3 Data

Source	S1 = FronkonGames dataset (Hugging Face); S2 = live Steam store + reviews.
Obtained	S1 download (Sep 2025 snapshot). S2: 1,000 random S1 games (random seed 7202), scraped 2026-10-14.
Legal	S1 MIT licence. S2: robots.txt allows /app/, /appreviews/; 1 req/1.5 s. No personal data.
Period	Released Jan 2023 – Dec 2024.
Volume	S1: all 9,094 Indie games with ≥10 reviews. S2: 996 with a live page.
Check	S1 vs S2: mean top-5 tag Jaccard 0.87; Spearman ρ of reviews 0.98.
Bias	Favours games that sold; reviews count buyers, not unmet demand; late-2024 games had less time.

4 Graph construction

Node	A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph.
Edge	Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation.
Resolution	Co-op, Online/Local Co-Op, Local Multiplayer merged into Multiplayer: 84% of Horror × Multiplayer games were also co-op, so splitting them hid crowding.
Weight	NPMI: co-occurrence vs chance, −1 to 1 (0 = chance). By raw count, 9 of the top-10 edges touch the 5 commonest tags; by NPMI, 0.
Window	Top 5, as shown on the store page. With all 20, the pair covers 375 games (mostly minor tags) and its reviews drop from 5.1× to 3.0×.
Excluded	"Indie" (genre filtered on); release-model, adult tags; >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration.
Pruning	Tag on ≥1.5% of games (136); edge: NPMI ≥ 0.10, ≥0.5% shared (45). 439 tags → **80 nodes, 177 edges**.

5 The picture



First-Person and Multiplayer share only Survival Horror. 14 of 80 tags; size = games; width = NPMI.

Selection rule: among pairs on ≥73 games (0.8%), keep those **not crowded** (games ≤ 1.5× chance, where chance = games with A × games with B ÷ 9,094) whose tags sit in **separate neighbourhoods** (≤20% shared neighbours); pick the one whose median reviews are highest relative to the median game (40).

Pair	Games (chance)	Crowding	Shared neighbours	NPMI	Median reviews
Pick: **First-Person × Multiplayer**	78 (71)	1.10×	8%	0.02	204 (5.1×)
Plan B: **Horror × Multiplayer**	114 (93)	1.23×	8%	0.05	117 (2.9×)
Crowded: **Building × Simulation**	74 (29)	2.54×	8%	0.19	136 (3.4×)

6 Findings

- **First-Person × Multiplayer** is near chance (NPMI 0.02, no edge), yet its median game gets 5.1× the reviews (permutation $p < 0.001$); 68% are horror, led by Lethal Company.
- 8 of the 10 highest-review pairs are crowded (>1.5× chance), e.g. Building × Simulation (2.54×).
- **Robust:** #1 at all 8 edge cutoffs; above the median game in 100/100 bootstraps. **Weak spot:** #1 in 59/100 (top 3 in 74), near the 73-game minimum.

7 Recommended action

- Week 0: pitch a first-person multiplayer horror game; plan B is Horror × Multiplayer.
- Weeks 1–6: prototype. Week 7: Coming Soon page, tagged First-Person and Multiplayer first.
- Before full production: re-run on 2025 releases; switch if crowding passes 1.5× or the pair leaves the top 3.

8 What would change my mind

"Not crowded" is against chance, not the 2023–24 co-op horror boom. If 2025 releases push it past 1.5×, I switch to the #3 pair, Pixel Graphics × RPG.